QIAGEN HMW Genomic DNA Extraction Modified Protocol (insect as input)

Last updated: Aug 13, 2025

Prepare:

1. Prepare 2n mL **Buffer QF** in tubes and heat up to 50 ℃ in a water bath.
2. Put **glycogen** on ice.
3. Prepare at least half a liter of **liquid nitrogen**.
4. Prepare n mortars and n pestles.
5. Prepare a bucket of ice and put samples on ice.
6. Prepare **G2+RNAse mix**: 2n mL **buffer G2** and 2n µL **RNAse** (100mg/mL), or 1959n µL **buffer G2** and 21n µL **RNAse** (10mg/mL) (final concentration of RNAse should be 100 µg/mL*). Leave on ice.
    a. *Note: *Note: The Qiagen Genomic Handbook recommends a final RNase concentration of 200 µg/mL. However, this is quite high, and the protocol is effective at half that concentration. If concerned about RNA contamination, you can extend the incubation time in step 7*
    b. For a 200 *µg/mL* concentration mix the following Prepare **G2+RNAse mix**: 2n mL **buffer G2** and 4n µL **RNAse** (100mg/mL), or 1959n µL **buffer G2** and 41n µL **RNAse** (10mg/mL) (final concentration of RNAse should be 100 µg/mL*).

Protocol:

1. Clean a ladle using <u>RNAse Away</u> and ethanol.
2. Pre-cool mortars and pestles using **LN**.
3. Grind the wasps with LN and transfer them into 2mL **DNA LoBind tubes**. <u>Do not cap the tubes.</u>
4. Repeat 1 & 2 until all samples have been grinded and transferred into 2mL **DNA LoBind tubes**.
5. After LN evaporates, add 2mL **G2+RNAse mix**.
6. Add 50-100µL **Proteinase K** or **Protease** provided by the kit.
7. Incubate for 2hr at 50℃ in the **ThermoMixer**. Gently mix the sample by inversion every 30 min.
    a. 10 min before incubation is done, let the centrifuge run for 10 min at 4℃ to lower the temperature.
    b. Note: A water bath can also be used, with the sample gently mixed by frequent inversion.
8. Gently vortex the sample for 10s and centrifuge at 5,000xg for 10 min at 4℃.

a. Get n 15mL tubes and put filters with racks onto the tubes.
b. Add 1mL Buffer QBT to each filter, and wait till empty by gravity flow.

9. Transfer the supernatant using wide-pore tips to new 2mL **DNA LoBind tubes**. <u>Be careful and avoid spilling.</u>
10. Centrifuge at 5,000xg for 1 min at 4℃. Make sure all debris is removed.
11. Transfer the supernatant using wide-pore tips to the filters. <u>Be careful and avoid spilling.</u> Wait till empty by gravity flow.
12. Wash the filters using 1mL of **Buffer QC** 3 times. Wait till empty by gravity flow.
13. Dry the tip of the column using Kimwipes to remove any residual liquid.
14. Transfer the filters to new 15mL tubes.
15. Add 1mL 50 ℃ **Buffer QF** to each filter 2 times, and wait till empty by gravity flow.
16. Split the samples into two 2mL **DNA LoBind tubes** using wide-pore tips.
17. In each tube, add 0.7 mL **isopropanol** and 1 µL **glycogen**.
18. Mix gently and incubate the samples at -20 ℃ for 1hr (optional).

    a. Sample can also be left overnight
19. Precipitate the DNA by centrifuging at 8,000xg for 15 min at 4℃. <u>Make sure the opening of the centrifuge tubes are pointing towards the center.</u>
a. Prepare 2n mL fresh 70% **ethanol** and put on ice.
20. Mark the DNA pellet on the tube and remove the supernatant carefully.
21. Wipe out the mark on the tube with ethanol.
22. Add 1mL 70% **ethanol** and mix gently.
23. Precipitate the DNA by centrifuging at 8,000xg for 10 min at 4℃. <u>Make sure the opening of the centrifuge tubes are pointing towards the center.</u>
24. Mark the DNA pellet on the tubes and remove the supernatant carefully.
25. Air-dry till all ethanol evaporates.
26. Add 25µL **EB Buffer** per tube (50 µL per sample) and incubate overnight at 4℃ to allow DNA to resuspend in solution. Alternatively you can also incubate at 50℃ for 1 hr, but some DNA will be lost.. <u>Never freeze HMW DNA.</u>
    <u>NOTE: Some EB buffers contain EDTA, which can interfere with downstream applications. Ideally, use EB buffer consisting of 10 mM Tris, pH 8.5, or opt for molecular-grade water instead.</u>
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Use this table to record the number of operations that have been done during the extraction:

Steps \ Samples				
12. Buffer QC wash (3)				
15. Buffer QF elute (2)				